

Major Project: AY 2025-26

College	RV College of Engineering, Bengaluru					
Department	Computer Science and Engineering					
Course:	B.E					
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Project Title	Machine Learning Based Parasite Recognition Software for Automated Detection of Plasmodium falciparum from Microscopic Images					
Project Type (Inhouse/Internship)	Inhouse					
Project Domain	Machine Learning and Deep Learning				TRL Level	5
					SDG	3
Under taken at	RVCE					
Online Course Enrolled Platform	NPTEL					
Course Title	Project Management					
Course Domain	Project Management					
Internal Guide			External Guide			
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			Designation			
Designation	Assistant Professor		Organization			
Stipend Amount (per month)						

Signature of Internal Guides
Date:

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SYNOPSIS

1. Project Overview

Malaria is a serious and potentially fatal disease caused by *Plasmodium* parasites, with *Plasmodium falciparum* being the most dangerous strain responsible for the majority of severe cases worldwide. The standard method of diagnosis involves microscopic examination of stained blood smear images by trained laboratory technicians. While this method is widely accepted, it is time-consuming, labor-intensive, and highly dependent on the expertise of the examiner. In addition, manual analysis is prone to human error, inconsistencies, and fatigue, especially when processing large numbers of samples.

In many developing and remote regions, the lack of skilled healthcare professionals further complicates timely diagnosis, leading to delays in treatment and increased risk of complications. The variability in interpretation between different experts can also affect the reliability of results. Moreover, traditional diagnostic methods are not easily scalable for large-scale screening, which is essential for effective disease control and prevention.

Therefore, there is a strong need for an automated and reliable system that can accurately detect malaria parasites from microscopic images. Such a system should be capable of analyzing large volumes of data efficiently while maintaining high accuracy and consistency. By leveraging advancements in machine learning and image processing, automated detection systems can reduce dependency on manual diagnosis and support faster, more accessible healthcare solutions.

2. Novelty / Innovation

The novelty of this project lies in the development of an end-to-end automated malaria detection system using a Convolutional Neural Network (CNN) that directly processes microscopic blood smear images without the need for manual feature extraction. Unlike traditional image processing and classical machine learning approaches that rely on handcrafted features, the proposed system automatically learns complex visual patterns such as parasite structures, texture variations, and color differences. This makes the model more robust, accurate, and adaptable to variations in image quality and staining conditions.

Another key innovation is the integration of the trained deep learning model into a Flask-based web application, enabling real-time prediction through a user-friendly interface. This bridges the gap between theoretical models and practical deployment by allowing users to upload images and obtain instant diagnostic results. The system can also be extended with explainable AI techniques such as Grad-CAM to visualize the regions influencing the prediction, enhancing transparency and trust. This combination of deep learning, automation, and real-time deployment makes the solution both practical and scalable for real-world healthcare applications.

3. Societal / Practical Contribution

The proposed system offers significant societal benefits by enabling faster and more accurate detection of malaria, which is crucial for timely treatment and prevention of severe complications. Early diagnosis plays a

vital role in reducing mortality rates, especially in regions where malaria is highly prevalent. By automating the detection process, the system minimizes human errors and inconsistencies associated with manual examination, ensuring more reliable and consistent diagnostic results.

From a practical perspective, the system reduces the dependency on highly skilled medical professionals, making it particularly useful in rural and resource-limited areas where access to trained pathologists is limited. The ability to analyze microscopic images quickly and efficiently allows healthcare providers to handle large volumes of samples, supporting mass screening programs and improving overall healthcare efficiency. Additionally, the use of a web-based interface makes the system easy to use and accessible, even for non-expert users.

Furthermore, the solution is cost-effective and scalable, as it is built using open-source technologies such as Python, TensorFlow, and Flask. It can be deployed on standard computing devices or extended to cloud-based platforms for wider accessibility. With further enhancements, the system can be integrated into mobile or telemedicine applications, contributing to modern, technology-driven healthcare systems. Overall, this project supports the goal of improving healthcare accessibility, efficiency, and quality through intelligent automation.

Reference:

Malaria Detection using CNN (Research Paper): <https://peerj.com/articles/4568/>

Deep Learning for Malaria Detection (IEEE): <https://ieeexplore.ieee.org/document/7961579>

World Health Organization Malaria Report: <https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report>

TensorFlow Documentation: <https://www.tensorflow.org>

Keras Documentation: <https://keras.io>

Flask Documentation: <https://flask.palletsprojects.com>

SDG Goals: <https://sdgs.un.org/goals>

TRL Levels: <https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/>